NATIONAL UNIVERSITY OF MODERN LANGUAGES

SOFTWARE QUALITY ENGINEERING LAB REPORT – 6

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COURSE: SOFTWARE QUALITY ENGINEERING

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Lab 6: Cost of Software Quality

Case Study: Online Banking Application

The software company developed an online banking application for a commercial bank. The system provides secure online banking services such as account management, fund transfers, bill payments, loan services, transaction history, notifications, and customer support integration.

Task 1: Categorization of Quality Engineering Activities

The following table classifies the quality engineering activities according to the Classic Model of Cost of Software Quality.

Quality Engineering Activity	**Cost ($)**	**Category**	**Reason**
Developer training / Workshops	5000	Prevention Cost	Training helps developers avoid defects during development.
Requirement analysis & validation	4000	Prevention Cost	Proper requirement analysis reduces requirement-related defects.
Software Design review	3500	Prevention Cost	Design reviews help identify defects before implementation.
Coding standards and guidelines enforcement	2500	Prevention Cost	Coding standards improve software quality and reduce errors.
Unit testing	4000	Appraisal Cost	Testing individual modules checks software quality.
Integration testing	3500	Appraisal Cost	Testing combined modules ensures proper integration.
Code inspections / Peer review	2000	Appraisal Cost	Peer reviews help identify defects before release.

Bug fixing during development	6000	Internal Failure Cost	Defects were found and corrected before delivery.
Rework after failed system testing	3000	Internal Failure Cost	Rework was required after testing failures before deployment.
Post release bug fixes / Patch deployment	8000	External Failure Cost	Defects were discovered after release by customers.
Customer support for bug related complaints	2000	External Failure Cost	Customer support handled complaints caused by software defects.

Task 2: Cost Calculation According to Categories

The following calculations show the total cost incurred under each category of the Classic Model of Cost of Software Quality.

Category	Activities Included	Total Cost ($)
Prevention Cost	Training, Requirement Analysis, Design Review, Coding Standards	15000
Appraisal Cost	Unit Testing, Integration Testing, Code Inspections	9500
Internal Failure Cost	Bug Fixing During Development, Rework After Failed Testing	9000
External Failure Cost	Post Release Bug Fixes, Customer Support Complaints	10000

Total Cost of Software Quality

Total Cost of Software Quality = Prevention Cost + Appraisal Cost + Internal Failure Cost + External Failure Cost

Total Cost = 15000 + 9500 + 9000 + 10000 = 43500$

Conclusion

The Online Banking Application project included several quality-related activities to improve software reliability, security, and performance. The costs were categorized using the Classic Model of Cost of Software Quality into Prevention, Appraisal, Internal Failure, and External Failure costs. The total cost of software quality for the project was calculated as 43,500$.